

Server Config – Arma Reforger

Category: [Arma Reforger/Support](#)

Arma Reforger servers uses [JSON](https://en.wikipedia.org/wiki/JSON) (<https://en.wikipedia.org/wiki/JSON>) configuration format. [Template #](#) and [Example #](#) are available at the bottom of this page.

Two **very important** aspects of dedicated server public hosting:



- Keep `fastValidation #` to `true`
- Limit max FPS with the `-maxFPS` startup parameter in order to save performance



- Parameters inside the JSON file are case-sensitive!
- Values are strings, unless mentioned otherwise.
- For Community config generators, see the following:
 - <https://arscfg.johandejong.dev/>
 - <https://armareforger.se/server-generator/>

Summary

Default ports to be opened for hosting:

Port	Protocol	Optional	Reason
2001	UDP		Public port #
17777			A2S port #
19999			RCON port #

Root

bindAddress

IP address to which the server socket will be bound. In most cases, this should be left empty. It can be used to restrict connections to a particular network interface. When left out or empty, `0.0.0.0` is used, which allows connections through any IP address.



- This entry should generally be left out of server config to use default value and should only be set by experienced users with specific network needs.
- IPv6 is **not** supported by Arma Reforger.

was ***gameHostBindAddress*** before 0.9.8.73

bindPort ___

number value, range 1..65535, default: 2001

UDP port to which the server socket will be bound.



This entry should generally be left out of server config so that it automatically uses same value as publicPort #. It should only be used by users with specific port forwarding needs.

was ***gameHostBindPort*** before 0.9.8.73

publicAddress ___

IP address registered in backend. This should be set to the public IP address to which clients can connect in order to reach the server (either IP of the server itself or IP of the machine that will forward data to the server). If the entry is missing or empty, then the public IP address will be automatically detected and used by the backend.



- This entry should generally be left out of server config to let the server automatically detect and use its public address used to reach it from internet.
- It is possible to use the "local" keyword to automatically detect the local address of the network card.
- IPv6 is **not** supported by Arma Reforger.

was ***gameHostRegisterBindAddress*** before 0.9.8.73

publicPort ___

number value, range 1..65535, default: 2001

UDP port registered in backend. If the server itself has a public IP address, this should be the same value as in bindPort #. Otherwise, this is the UDP port that is forwarded to the server.

was ***gameHostRegisterPort*** before 0.9.8.73



a2s ___

0.9.9.31

Steam Query protocol definition - see the a2s # section below.

Before  a2s settings were:



- a2sQueryEnabled - bool value, default: false
- steamQueryPort - number value, range 1..65535, default: 17777

rcon

1.0.0

The RCON Protocol is a UDP-based communication protocol that allows console commands to be issued to the server via a **R**emote **C**ONsole, or RCON - see the rcon # section below.

game

Define the server's settings - see the game # section below.



Only **one** scenario can be defined - Arma Reforger does **not** allow for mission rotation as of v0.9.8.

operating

0.9.7

Define various server settings - see the operating # section below.

a2s _____

0.9.9

0.9.9.31

See A2S server queries (https://developer.valvesoftware.com/wiki/Server_queries).

address

required

IP address to which A2S socket will be bound. It can be used to restrict A2S queries to a particular network interface.

port

number value, range 1..65535, default: 17777

Change Steam Query UDP port on which game listens to A2S requests

(Optional)

See [RCON protocol \(https://www.battleye.com/downloads/BERConProtocol.txt\)](https://www.battleye.com/downloads/BERConProtocol.txt).

address

required

IP address to which the RCON socket will be bound. It can be used to restrict connection to a particular network interface.

port

number value, range 1..65535, default 19999

RCON protocol port on which the game listens

password

string

The password:



- is required for RCON to start
- does **not** support spaces
- must be at least **3** characters long

maxClients

number value, range 1..16, default 16

The maximum number of clients that can connect to RCON at the same time.



RCON lacks client disconnect support, so when a user exits the RCON client, the server maintains the connection for an additional 30 seconds before a timeout removes the user from the list.

permission

string

Permission for all RCON clients. Possible values:

Key	Value
admin	The admin can perform any command.
monitor	The monitor can only perform commands which do not change the server's state.

blacklist

array value, default []

A list of commands excluded from execution.

whitelist

array value, default []

If defined, it specifies the list of commands that can be executed, and no other command is allowed.

game

name

length 0..100 characters

password

length 0..x characters

Password required to join the server.

passwordAdmin

length: 0..x characters

Defines the server's admin password, allows a server administrator to login and control the server, to access this either open the chat input box by pressing **C** in the lobby or **Return ↵** in-game followed by: #login [the admin password]



This password does not support spaces in it.

was root/passwordAdmin before 0.9.8.73

admins

0.9.9.31

array value of **IdentityIds** and/or steamIds

List players as server admins - they can be checked in script using `BackendApi.IsListedServerAdmin(int playerId)` regardless of their logged-in state.



Listed users can `#login` without a password.



1.3.1

- list is now limited to only 20 unique IDs
- listed admins can use priority queue when joining server
 - they will have their own queue which have priority before normal players when joining the server
 - priority queue works only with admins which are specified by **IdentityIds**
 - **admins specified by their SteamId are not currently supported for priority queue**

scenarioId

The scenario's `.conf` file path is defined here. See the [listScenarios](#) startup parameter to list available scenarios and obtain their `.conf` file path.



1.6.1

Show Official Scenarios (1.6.0)



1.3.1

Show Official Scenarios (1.3.0)



1.2.1

Show Official Scenarios (1.2.1)



1.1.1

Show Official Scenarios (1.1.0)



1.0.1

Show Official Scenarios (1.0.0)



0.9.1

Show Official Scenarios (0.9.9)

maxPlayers

number value, range 1..128, default: 64

Set the maximum amount of players on the server.

was *playerCountLimit* before 0.9.8.73

visible

bool value, default: true (since 0.9.8.73)

Set the visibility of the server in the Server Browser.



0.9.9

crossPlatform

bool value, default: false

Accept all platforms if set to true, use [supportedPlatforms](#) # if set to false.



It is recommended to use **crossPlatform** only, leaving supportedPlatforms # undefined.

supportedPlatforms

array value, default: ["PLATFORM_PC"]

Define the platforms which the server accepts, allowing crossplay. Possible values:

Key	Value
PLATFORM_PC	PC
PLATFORM_XBL	Xbox console
PLATFORM_PSN	PlayStation console (since Arma Reforger 1.2.1.169)

It is recommended to use crossPlatform # instead and leave this property undefined:



crossPlatform = true;

crossPlatform = false;

is
equivalent
to

supportedPlatforms = ["PLATFORM_PC", "PLATFORM_XBL", "PLATFORM_PSN"];

supportedPlatforms = ["PLATFORM_PC"];

These two being the only valid combinations.



was *supportedGameClientTypes* before 0.9.8.73

gameProperties

Define the scenario's settings - see the gameProperties # section below.

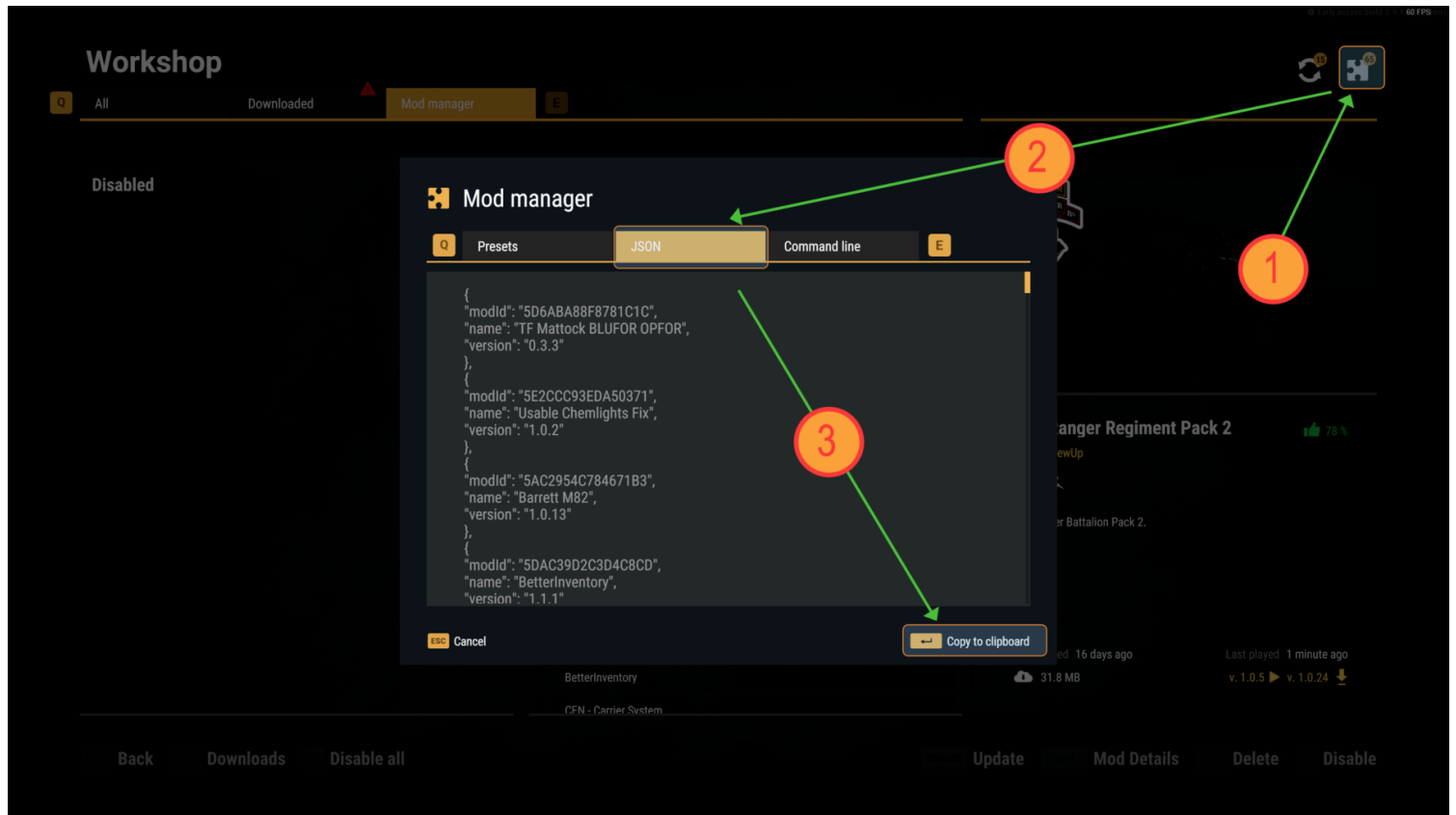
1.2.1 modsRequiredByDefault

bool value, default: true

Overrides default value for required # for all mods.

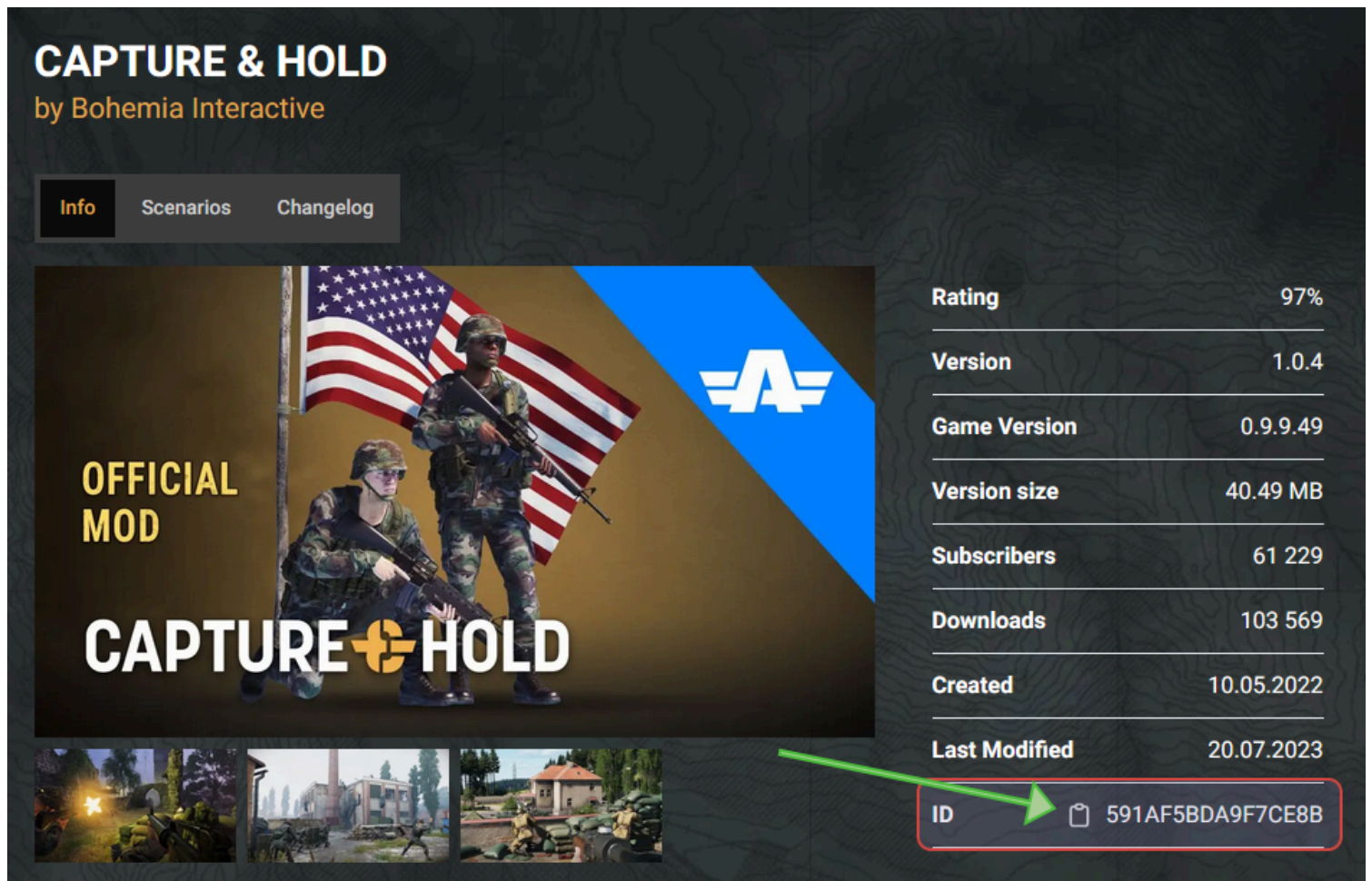
mods

The list of mods required by the client that will automatically be downloaded and activated on join. It can be created by activating wanted mods in the Workshop and then, via the **Mod manager**, result can be converted to ready to be used string after navigating to **JSON** tab and then clicking on the **Copy to clipboard** button.

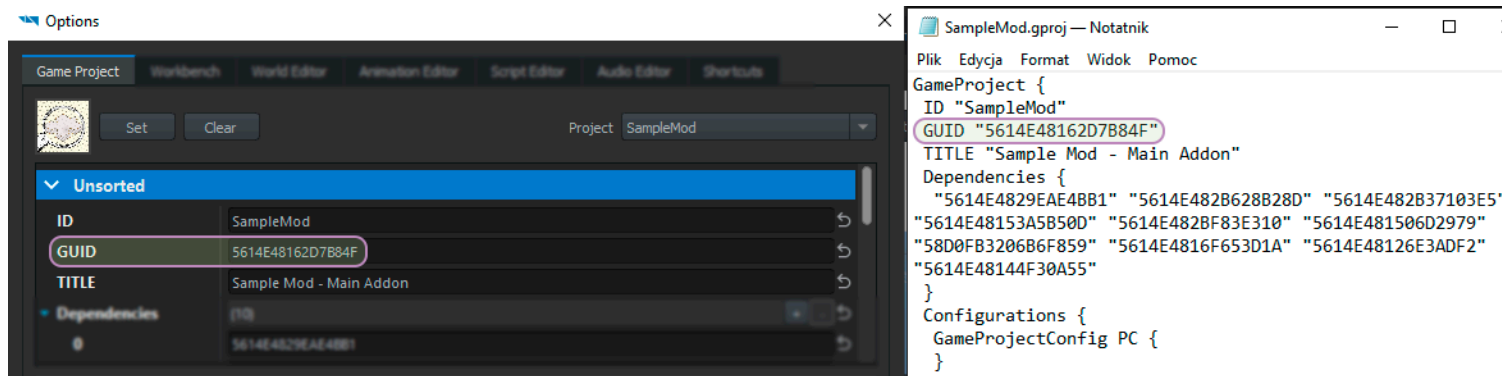


modID __

GUID of the mod. Single GUID can be obtained from Workshop webpage



Alternatively you can grab it from Workbench options when mod is running or directly from gproj file



name _____

This parameter does not do anything and is only used as sort of comment, with human readable name of the mod.

version _____

 The `version` mod parameter is optional. If it missing, the latest mod version will be used.

 **required** _____

bool value, default: true or value of `modsRequiredByDefault` #

Is optional parameter to specify if this addon is required for server to start. If set to false then addon will be automatically removed from list with warning in logs if it cannot be for some reason downloaded from the Workshop.

gameProperties

serverMaxViewDistance _____

number value, range 500..10000, default: 1600

serverMinGrassDistance _____

number value, range 0 / 50..150, default: 0

Minimum grass distance in meters. If set to 0 no distance is forced upon clients.

fastValidation _____

bool value, default: true (since 0.9.6)

Validation of map entities and components loaded on client when it joins, ensuring things match with initial server state.

- true (enabled) - minimum information required to make sure data matches is exchanged between client. When a mismatch occurs, no additional information will be available for determining where client and server states start to differ. All servers that expect clients to connect over internet should have fast validation enabled.
- false (disabled) - extra data for every replicated entity and component in the map will be transferred when new client connects to the server. When a mismatch occurs, it is possible to point at particular entity or component where things start to differ. When developing locally (ie. both server and client run

on the same machine), it is fine to disable fast validation to more easily pin point source of the problem.



Always set this value to `true` for a public server!

networkViewDistance

number value, range 500..5000, default: 1500

Maximum network streaming range of replicated entities.

battlEye

bool value, default: true (since 0.9.8.73)

true to enable BattlEye, false to disable it.

disableThirdPerson

bool value, default: false

Force clients to use the first-person view.

VONDisableUI

bool value, default: false

Force clients to not have VON (Voice Over Network) UI.

VONDisableDirectSpeechUI

bool value, default: false

Force clients to not have VON (Voice Over Network) Direct Speech UI.



VONCanTransmitCrossFaction

bool value, default: false

Option to allow players to transmit on other factions radios. true is allow to communicate, false is listen-only

missionHeader

This property overwrites the scenario's header, allowing to e.g change the scenario's displayed name:

```
"missionHeader": {  
  "m_sName": "My Very Own Hosted Conflict",  
  "m_sDetails": "If you teamkill or commit war crimes, you WILL be banned!",  
  "m_iStartingHours": 7,  
  "m_iStartingMinutes": 30,
```

```
}  
    "m_bRandomStartingWeather": true  
}
```

Generic scenario properties (from SCR_MissionHeader ([enfusion://ScriptEditor/scripts/Game/Mission/SCR_MissionHeader.c;1](#))): [Show text](#)

Conflict-specific properties (from SCR_MissionHeaderCampaign ([enfusion://ScriptEditor/scripts/Game/Mission/SCR_MissionHeaderCampaign.c;1](#))): [Show text](#)

persistence

1.6.0

Configuration values for the Persistence System that can be optionally defined. The system is set up to automatically work for most use cases by default. [Show possible options](#)

To disable persistence entirely, set the enabled save types in `game->gameProperties->missionHeader` to 0 as shown below:

```
"missionHeader": {  
    "m_eSaveTypes": 0  
}
```

autoSaveInterval

number value, range 0..60, default: 10

Interval (in minutes) at which automatic saves are created, if possible for the current mission. Disabled if set to 0.

saveRetention

number value, range 1..128, default: 10

Number of save points to keep for the current mission.

loadSessionSave

boolean value, true/false, default: true

Automatically load the latest available save point on first startup.

keepSessionSave

boolean value, true/false, default: false

Keep the playthrough save points after the mission is finished.

hiveId

number value, range 0..16383, default: 0

Number identifying each hive (server) when they share the same persistence database. Used for separating UUIDs for save game data.

databases

(Case-insensitive) named objects representing an override of the system configuration provided databases or the addition of a new one. Optional child properties:

- "preset" - refers to database config presets that are defined in game/mod data. Full resource name.
- "options" - a complex object which corresponds to the individual database type of the preset. Usually, a key-value collection

storages ___

(Case-insensitive) named objects representing an override of the system configuration-provided storages. Optional child properties:

- "database" - changes the storage to use a different database than the default configured. Must match the name given in the databases object or "main".

operating 0.9.7 ___

lobbyPlayerSynchronise ___

bool value, default: true (since 0.9.8.73)

If enabled, the list of player identities present on server is sent to the GameAPI along with the server's heartbeat.



This setting fixes the discrepancy between the real and reported number of players on the server.

disableCrashReporter 0.9.8 ___

bool value, default: false

If enabled, the automatic server-side Crash Report is disabled. Has the same effect as -disableCrashReporter.

disableNavmeshStreaming 1.0.0 ___

 bool value, default: false

 array value, default undefined

A disabled navmesh streaming loads the entire navmesh in memory - this provides slightly better server performance and reaction times of moving AIs at the cost of a bigger memory footprint (up to hundreds of MB depending on the terrain) - see 2024-07-07 Modding Update (<https://reforger.armaplatform.com/news/modding-update-june-7-2024>)

-  1.0.0 If set to true, navmesh streaming is disabled on all navmesh components
-  1.2.0 If not provided, no navmesh streaming is disabled; if provided empty, streaming of all navmeshes is disabled; if filled, only disables provided navmeshes's streaming

was **bool** before 1.2.0

disableServerShutdown 0.9.8 _____

bool value, default: false

If enabled, the server will not automatically shutdown if connection to backend is lost. Related to room requests errors - other causes like corrupted config will still shutdown the server.

disableAI 1.1.0 _____

bool value, default: false

If enabled, the server will prevent initialization and ticking of AIWorld and its components. Will completely disable AI functionality on the server.

playerSaveTime 0.9.8 _____

number value, default: 120

Default period in seconds for saving players for both Online and Local storage (player save can still be requested on demand).

aiLimit 0.9.8 _____

0.9.8.64

number value, default: -1

Sets the top limit of AIs. No system will be able to spawn any AIs when this ceiling is reached (through `aiWorld.CanAICharacterBeAdded()`)

A negative number is not considered as valid value and is then ignored - limit is not applied.

This param is overridden by the `-aiLimit` startup parameter.

slotReservationTimeout 0.9.9 _____

0.9.9.31

number value, range 5..300, default 60

Sets the duration (in seconds) for how long will the backend and server reserve a slot for kicked player. It is considered disabled when set to the minimal value, the value being the same as for a normal disconnect.

It can be used in scripts *via* Game Mode events:

```
OnPlayerAuditTimeouted(int iPlayerID) // invoked when player did not recor
```

```
OnPlayerAuditRevived(int iPlayerID) // invoked when player successfully
```



Reservation works only for **replication** kicks.

joinQueue

1.2.1

1.2.1.66

Configuration for player join queue to the server

maxSize

1.2.1

number value, range 0..50, default 0 (disabled)

Sets maximum size of how many people can be at one time in join queue to the server.

Template

```
{
  "bindAddress": "",
  "bindPort": 0,
  "publicAddress": "",
  "publicPort": 0,
  "a2s": {
    "address": "",
    "port": 0
  },
  "rcon": {
    "address": "",
    "port": 0,
    "password": "",
    "permission": "monitor",
    "blacklist": [],
    "whitelist": []
  },
  "game": {
    "name": "",
    "password": "",
    "passwordAdmin": "",
    "admins" : [],
    "scenarioId": "",
    "maxPlayers": 0,
    "visible": true,
    "gameProperties": {
      "serverMaxViewDistance": 1600,
      "serverMinGrassDistance": 0,
      "networkViewDistance": 1500,
```

```
        "disableThirdPerson": false,  
        "fastValidation": true,  
        "battlEye": true,  
        "VONDisableUI": false,  
        "VONDisableDirectSpeechUI": false,  
        "VONCanTransmitCrossFaction": false  
    },  
    "mods": []  
}  
}
```

Example

Show Example Configuration

Retrieved from "https://community.bistudio.com/wiki?title=Arma_Reforger:Server_Config&oldid=378701"